NLU Assignment 2

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Code Repository:
Github

1 Introduction

This report presents the implementation and analysis of two problems:

- **Problem 1:** Learning word embeddings using Word2Vec (CBOW and Skip-gram)
- **Problem 2:** Neural text generation using RNN, BLSTM, and Attention models

The goal is to analyze semantic representation and evaluate generative capabilities across architectures.

2 Problem 1: Word Embedding Learning

2.1 Dataset Collection and Preprocessing

A web crawler was implemented to collect data from IIT Jodhpur-related sources. The pipeline includes:

- URL crawling (depth = 2, max pages = 120)
- HTML/PDF parsing
- Boilerplate removal
- Tokenization and cleaning

2.2 Dataset Statistics

- Corpus Size: 254 KB
- Total documents: 118
- Total tokens: 37,224
- Vocabulary size: 4,315
- Sources used: 118
- Seed URLs: 52

2.3 Frequent Words

Top words include: *of, department, the, engineering, institute, research, technology*

Observation: The dataset is highly domain-specific with strong academic and administrative bias.

2.4 Word Cloud Analysis

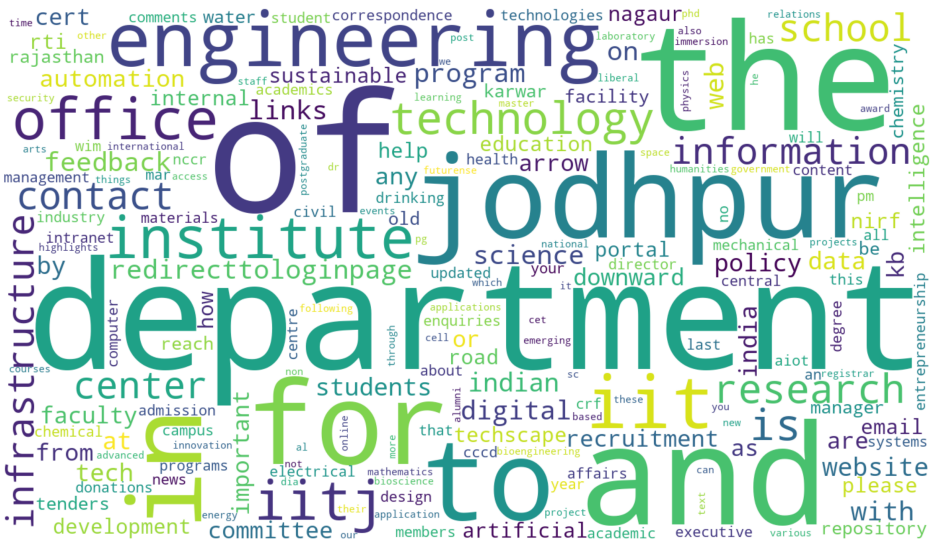

Figure 1: Word Cloud of Corpus

The word cloud confirms dominance of institutional and research-related terms.

2.5 Word2Vec Training

Models were trained using a grid search over:

- Dimensions: 100, 200, 300
- Window sizes: 3, 5, 8
- Negative samples: 5, 10

2.6 Best Models

- **CBOW:** dim=200, window=3, negative=5
- **Skip-gram:** dim=100, window=3, negative=5

2.7 Semantic Analysis

Semantic evaluation was performed using two approaches:

- Nearest neighbor analysis (cosine similarity)
- Analogy tasks of the form $a : b :: c :?$

CBOW Results:

- **Nearest Neighbors:**
  - research → crf, economics, industrial
  - student → donate, council (noisy associations)
  - phd → advanced, curriculum, doctoral
  - exam → [out-of-vocabulary]

- **Analogy Tasks:**
  - ug:btech :: pg:? → jee, jam (weak semantic alignment)
  - btech:undergraduate :: mtech:? → graduate, doctoral
  - research:lab :: teaching:? → website, unrelated tokens

Observations (CBOW):

- Produces **smoother embeddings** due to context averaging
- Suffers from **semantic dilution**, leading to noisy neighbors
- Performs poorly on analogy tasks requiring precise vector arithmetic

Skip-gram Results:

- **Nearest Neighbors:**
  - research → analyzer, spectrometer, performance
  - student → life, wellbeing, intake
  - phd → masters, semester, responsibility
  - exam → [out-of-vocabulary]
- **Analogy Tasks:**
  - ug:btech :: pg:? → diploma, certificate (strong alignment)
  - btech:undergraduate :: mtech:? → graduate, curriculum
  - research:lab :: teaching:? → structure, reporting (partially meaningful)

Observations (Skip-gram):

- Learns **sharper and more discriminative embeddings**
- Captures **semantic and functional relationships** more effectively
- Performs significantly better on analogy reasoning

Overall Comparison:

- Skip-gram outperforms CBOW in both **neighbor quality** and **analogy tasks**
- CBOW is computationally efficient but loses fine-grained semantics
- Both models suffer from **out-of-vocabulary issues**, indicating vocabulary limitations in the dataset

2.8 PCA Visualization

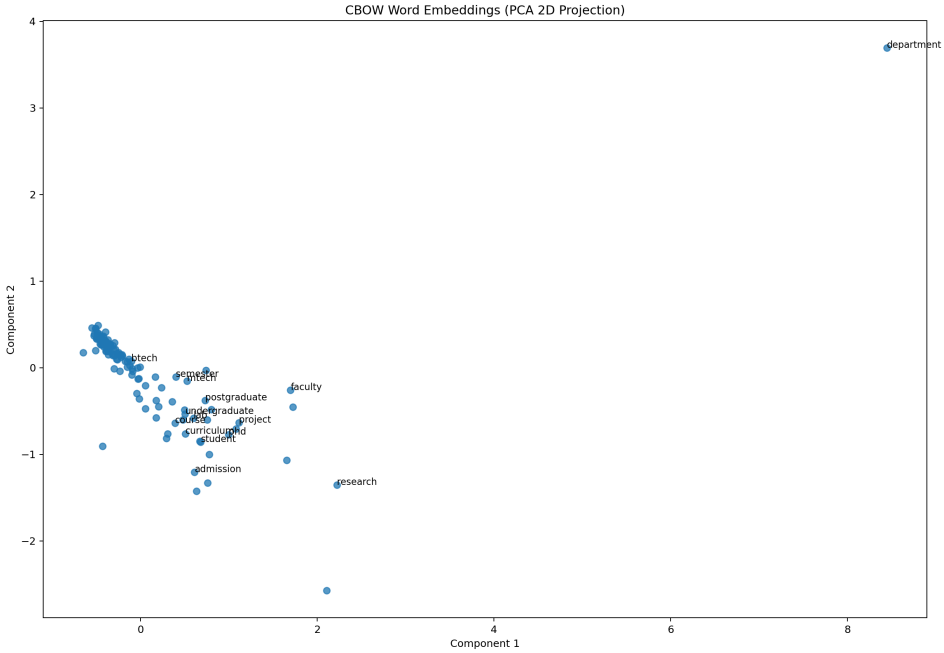

Figure 2: CBOW Embeddings (PCA)

Observation: Dense clustering with weak separation.

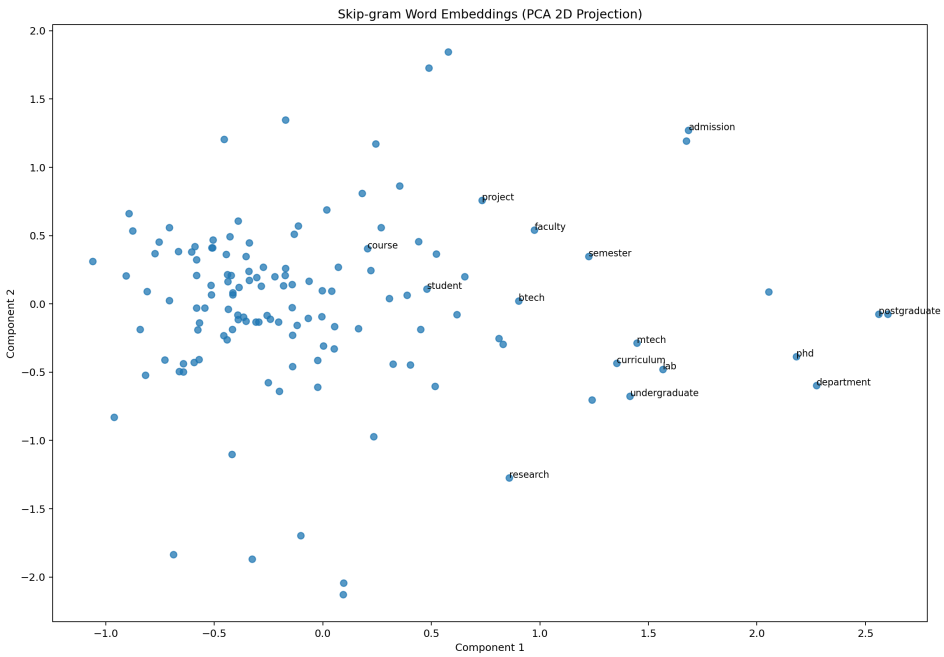

Figure 3: Skip-gram Embeddings (PCA)

Observation: Better spread and clearer semantic grouping.

2.9 Key Takeaways

- Skip-gram outperforms CBOW in semantic tasks

- CBOW is faster but less expressive
- Dataset noise impacts embedding quality

3 Problem 2: Neural Text Generation

3.1 Models Implemented

- RNN
- BLSTM
- RNN with Attention

3.2 Evaluation Metrics

- **Novelty:** Fraction of generated samples not in training data
- **Diversity:** Fraction of unique generated outputs

3.3 Results

Model	Parameters	Novelty	Diversity
RNN	146,974	0.01	0.60
BLSTM	275,742	1.00	0.826
Attention	147,231	0.926	0.024

Table 1: Model Performance Comparison

3.4 Diversity Comparison

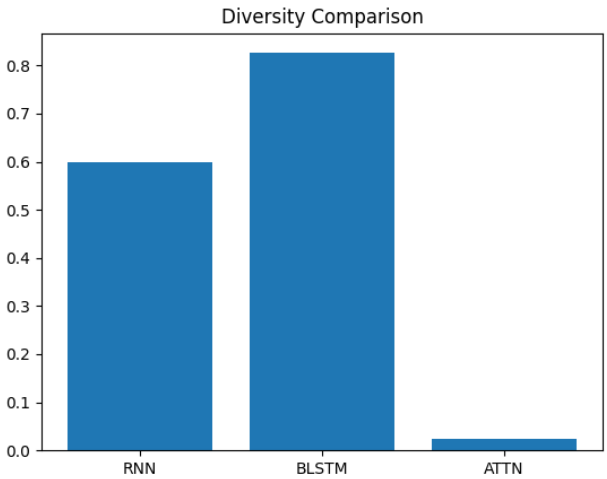

Figure 4: Diversity Comparison

3.5 Novelty Comparison

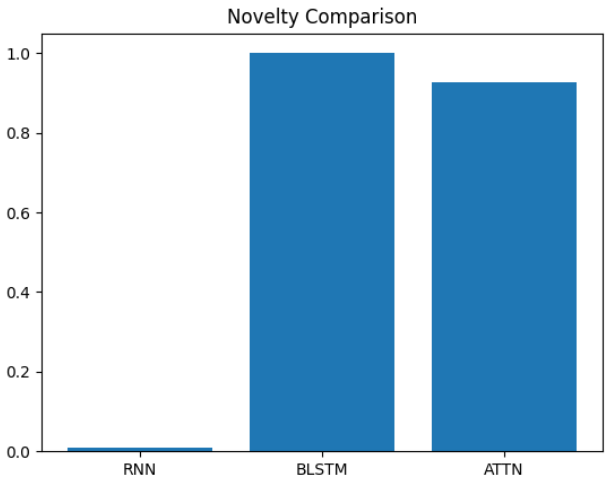

Figure 5: Novelty Comparison

3.6 Model Analysis

RNN:

- Low novelty (memorization)
- Moderate diversity

BLSTM:

- Best performance overall
- High novelty and diversity

Attention Model:

- High novelty
- Extremely low diversity (mode collapse)

3.7 Key Takeaways

- BLSTM performs best across all metrics
- Attention suffers from repetition issues
- RNN lacks expressive power

4 Conclusion

This work demonstrates:

- Skip-gram produces better semantic embeddings than CBOW
- BLSTM is superior for sequence generation

- Model design significantly impacts diversity and novelty

Future improvements include:

- Better preprocessing to reduce noise
- Hyperparameter tuning for attention models
- Larger and more diverse datasets